

Miao Chen

Graduate Student in Physics | School of Physics and Astronomy, Beijing Normal University

✉ Email: miao chen@mail.bnu.edu.cn  ORCID  Google Scholar  Homepage

Research Interests

Non-equilibrium statistical physics; phase transitions; information thermodynamics

Education

M.Sc. in Theoretical Physics, Beijing Normal University, Beijing, China Sep. 2024–present
Advisor: Prof. Yu-Han Ma GPA: 3.8/4

B.Sc. in Physics, Jiangxi Normal University, Nanchang, China Sep. 2020–Jun. 2024
Advisor: Prof. Hui-Huang Chen GPA: 90.03/100 Honor: Outstanding Graduate

Publications

- Miao Chen***, Xiu-Hua Zhao*, and Yu-Han Ma[†], “Coercivity Landscape Characterizes Dynamic Hysteresis,” *Physical Review Letters* **136**, 117102 (2026), *Editors’ Suggestion*.
- Miao Chen**, Xiu-Hua Zhao[†], and Yu-Han Ma[†], “Finite-time and finite-size scalings of coercivity in dynamic hysteresis,” *Physical Review E* **113**, 034124 (2026).
- Wanyan Chen*, **Miao Chen***, and Yu-Han Ma[†], “Finite-Time Thermodynamics of an Autonomous Information Machine,” *arXiv:2604.15953* (2026).
- Miao Chen** and Hui-Huang Chen[†], “Rényi entanglement asymmetry in (1+1)-dimensional conformal field theories,” *Physical Review D* **109**, 065009 (2024).
- Miao Chen** and Qin Guo, “Theoretical Study and Visualization of the Winding Problem,” *College Physics* **43**(4), 65–68, 80 (2024). [in Chinese]

Research Experience

School of Physics and Astronomy, Beijing Normal University Dec. 2023–present
Advisor: Prof. Yu-Han Ma

- Developed a coercivity-landscape framework to characterize finite-time and finite-size scaling behaviors in dynamic hysteresis using renormalization techniques.
- Performed Kinetic Monte Carlo simulations of the two-dimensional Ising model in driven systems.
- Studied finite-time thermodynamics and performance trade-offs in information machines.

Department of Physics, Jiangxi Normal University Feb. 2023–Jun. 2024
Advisor: Prof. Hui-Huang Chen

- Investigated symmetry breaking and entanglement-related quantities in conformal field theories.
- Implemented numerical tests of CFT predictions using exactly solvable lattice models.

Teaching Experience

- Teaching Assistant, College Physics at BNU, 2025.
- Teaching Assistant, Advanced Mathematics at JXNU, 2021–2023.

Conferences

International Conference on Statistical Physics (29th) Florence, Italy.
Poster: Coercivity Panorama of Dynamic Hysteresis Jul. 13–18, 2025

Progress in Quantum Machine Learning (2026) Beijing, China.
Poster: Coercivity Landscape Characterizes Dynamic Hysteresis May 10, 2026

National Conference on Magnetism Theory (20th) Hangzhou, China.
Talk: Coercivity Landscape Characterizes Dynamic Hysteresis May 22–25, 2026